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Clinical Investigation: Central Nervous System Tumor

Plaque Brachytherapy for Uveal Melanoma: A Vision Prognostication Model

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Summary

A visual prognostication model for predicting visual acuity after treatment with I-125 or Ru-106 plaque brachytherapy for uveal melanoma was generated, based on the experience of treating 189 patients. The resulting nomogram provides a means to predict vision loss at 3 years after reatment, which may have an impact on patient selection and treatment counseling.

Purpose: To generate a vision prognostication model after plaque brachytherapy for uveal melanoma.

Methods and Materials: All patients with primary single ciliary body or choroidal melanoma treated with iodine-125 or ruthenium-106 plaque brachytherapy between January 1, 2005, and June 30, 2010, were included. The primary endpoint was loss of visual acuity. Only patients with initial visual acuity better than or equal to 20/50 were used to evaluate visual acuity worse than 20/50 at the end of the study, and only patients with initial visual acuity better than or equal to 20/200 were used to evaluate visual acuity worse than 20/200 at the end of the study. Factors analyzed were sex, age, cataracts, diabetes, tumor size (basal dimension and apical height), tumor location, and radiation dose to the tumor apex, fovea, and optic disc. Univariate and multivariable Cox proportional hazards were used to determine the influence of baseline patient factors on vision loss. Kaplan-Meier curves (log rank analysis) were used to estimate freedom from vision loss.

Results: Of 189 patients, 92% (174) were alive as of February 1, 2011. At presentation, visual acuity was better than or equal to 20/50 and better than or equal to 20/200 in 108 and 173 patients, respectively. Of these patients, 44.4% (48) had post-treatment visual acuity of worse than 20/50 and 25.4% (44) had post-treatment visual acuity worse than 20/200. By multivariable analysis, increased age (hazard ratio [HR] of 1.01 [1.00-1.03], $P = .05$), increase in tumor height (HR of 1.35 [1.22-1.48], $P < .001$), and a greater total dose to the fovea (HR of 1.01 [1.00-1.01], $P < .001$) were predictive of vision loss. This information was used to develop a nomogram predictive of vision loss.

Conclusions: By providing a means to predict vision loss at 3 years after treatment, our vision prognostication model can be an important tool for patient selection and treatment counseling.

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